

**MAHARASHTRA STATE BOARD OF TECHNICAL EDUCATION,
MUMBAI**

Industrial Training Report of

**RAINFALL PREDICTION USING LONG SHORT-TERM MEMORY
(LSTM) NEURAL NETWORKS**

Submitted By

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(2026 - 27)

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CERTIFICATE

Certified that the Industrial Training Report entitled

**RAINFALL PREDICTION USING LONG SHORT-TERM MEMORY (LSTM)
NEURAL NETWORKS**

has been successfully completed by

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As partial fulfillment of the Diploma in Computer Technology under the Maharashtra State Board of Technical Education, Mumbai, during the academic year 2026-27.

The said work has been assessed by us and we are satisfied that the same is up to the standard envisaged for the level of the course, and that the said work may be presented to the external examiner.

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ABSTRACT

This report presents the design, development and testing of a rainfall prediction system built using Long Short-Term Memory (LSTM) neural networks. Thirty-five years of daily weather observations for six Indian cities were obtained from the Open-Meteo ERA5 reanalysis archive. A two-headed LSTM was trained for each city to predict, one day ahead, both the probability of rainfall and the quantity of rainfall in millimetres. The models were validated against persistence and climatology baselines using strictly chronological data splits. The system achieves a classification F1 score of 0.931 and a coefficient of determination of 0.462 for Mumbai, outperforming both baselines. A web application was developed using Streamlit to deliver real-time forecasts.

Keywords : Machine Learning, Deep Learning, LSTM, Recurrent Neural Network, Time Series Forecasting, Rainfall Prediction, PyTorch, Streamlit, Open-Meteo, ERA5.

ACKNOWLEDGEMENT

I express my gratitude to Dr. V. R. Rathi, Principal, P. Dr. V. V. P. Institute of Technology and Engineering, Pravaranagar, and [Name of HOD], Head of the Computer Technology Department, for their kind help and co-operation.

It is my privilege to acknowledge with a deep sense of gratitude my Mentor [Name of Mentor] for his valuable suggestions and expert guidance throughout my course of study, and for the timely help given to me in the completion of this industrial training report.

I am highly obliged to the entire staff of the Computer Technology Department for their kind help and co-operation. I would also like to express my appreciation and thanks to all my colleagues and family members who knowingly or unknowingly have assisted and encouraged me throughout my journey.

I am glad to express my sentiments of gratitude to all who rendered their valuable help in the successful completion of this report.

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CHAPTER 1

ORGANIZATION STRUCTURE OF INDUSTRY AND GENERAL LAYOUT

1.1 INTRODUCTION

The industrial training was carried out at [Name of the Organization], located at [Address of the Organization]. The organization operates in the Information Technology sector and is engaged in the design and development of software products, with a dedicated practice in data science and machine learning. The training was undertaken in the Software Development and Data Science division, where the trainee was assigned the development of a rainfall prediction system based on deep learning.

1.2 ORGANIZATIONAL STRUCTURE

The organization follows a functional hierarchy in which technical work is organized into project teams. Each team is self-contained and responsible for a product or module from requirement analysis through to deployment and maintenance. The hierarchy observed during the training period is summarized in Table 1.1.

Table 1.1 : Organizational hierarchy and responsibilities

Level	Designation	Principal Responsibilities
1	Director / Chief Executive Officer	Overall strategy, business development and client relations
2	Project Manager	Project planning, scheduling, resource allocation and client reporting
3	Team Lead / Technical Lead	Technical design decisions, code review and mentoring of team members
4	Senior Software Engineer	Module design, implementation of core components and unit testing
5	Software Engineer / Data Scientist	Implementation, data preparation, model development and documentation
6	Quality Assurance Engineer	Test planning, functional testing, defect reporting and verification
7	System Administrator	Server maintenance, network administration, backup and data security
8	Trainee / Intern	Assigned development tasks under the supervision of a mentor

1.3 DEPARTMENT IN WHICH TRAINING WAS UNDERTAKEN

The trainee was attached to the Data Science and Machine Learning team. This team is responsible for building predictive models from historical data, for validating those models against agreed accuracy criteria, and for delivering the resulting models to end users through web applications or application programming interfaces. The team works closely with the web development team for the user interface and with the system administration team for deployment.

1.4 GENERAL LAYOUT OF THE WORKPLACE

The working premises are arranged as an open-plan development floor. The layout is designed so that project teams sit together, allowing informal discussion while keeping noise away from areas that require concentration. The principal areas of the workplace are described below.

1.4.1 Development area

An open workspace containing individual workstations, each equipped with a desktop or laptop computer, a secondary display, and a network point. Workstations are grouped by project team.

1.4.2 Server and network room

A separate, air-conditioned and access-controlled room housing the servers, network switches, router and uninterruptible power supply units. Entry is restricted to authorized personnel of the system administration team.

1.4.3 Meeting and discussion rooms

Enclosed rooms fitted with a projector or large display, used for daily stand-up meetings, design discussions, code reviews and client calls.

1.4.4 Reception, administration and common facilities

The reception and administrative offices are located near the entrance. Common facilities include a pantry, rest rooms and a first-aid point. Fire extinguishers and clearly marked emergency exits are provided on each floor, and the emergency assembly point is marked outside the building.

CHAPTER 2

INTRODUCTION TO THE INDUSTRY / ORGANIZATION

2.1 HISTORY AND BACKGROUND

[Name of the Organization] was established in the year [Year of Establishment] and has since been engaged in providing software development and information technology services. Over the years the organization has expanded its scope from conventional application development to include data analytics, machine learning and cloud-based solutions. The organization serves clients in domains such as education, healthcare, agriculture, retail and public services.

2.2 PRODUCTS AND SERVICES

The principal products and services offered by the organization are listed below. The rainfall prediction system developed during this training falls under the machine learning and data analytics service line.

- Custom web application development using modern frameworks.
- Mobile application development for the Android and iOS platforms.
- Data analytics, reporting and business intelligence dashboards.
- Machine learning and deep learning model development and deployment.
- Application programming interface design and systems integration.
- Cloud deployment, hosting, and ongoing technical support and maintenance.
- Software testing and quality assurance services.

2.3 TURNOVER AND MANPOWER

The approximate scale of the organization during the training period is given in Table 2.1. These figures were provided by the administrative department for the purpose of this report.

Table 2.1 : Organizational profile

Particulars	Details
Name of the organization	[Name of the Organization]
Year of establishment	[Year]
Nature of business	Information Technology - software development and data science
Address	[Address of the Organization]
Annual turnover (approximate)	[Turnover]
Total number of employees	[Number of Employees]
Technical staff strength	[Number]
Working hours	[e.g. 9:30 a.m. to 6:30 p.m., Monday to Saturday]
Training duration	[Start Date] to [End Date]

2.4 VISION AND MISSION

The stated vision of the organization is to deliver reliable and cost-effective software solutions that create measurable value for its clients. Its mission is to apply current technology, disciplined engineering practice and continuous learning to solve practical problems, while providing a supportive environment in which engineers and trainees can develop professionally.

2.5 DOMAIN OF THE TRAINING ASSIGNMENT

The assignment allotted to the trainee lies in the field of weather forecasting using machine learning. Rainfall is one of the most important meteorological variables in India, since agriculture, water resource management, urban drainage and disaster preparedness all depend directly upon it. Conventional numerical weather prediction requires the solution of complex physical equations on high-performance computers, which is expensive and not always available at a local scale. Machine learning offers an alternative approach in which a model

learns the statistical relationship between past weather observations and future rainfall directly from historical records.

Rainfall is a sequential phenomenon: the weather on any given day depends strongly on the conditions of the preceding days. The Long Short-Term Memory network, a specialized form of recurrent neural network, is designed for exactly this class of problem, and was therefore selected as the modelling technique for this assignment.

CHAPTER 3

MAJOR EQUIPMENT, HARDWARE AND SOFTWARE USED

3.1 INTRODUCTION

Being a software organization, the equipment used consists principally of computing hardware, networking infrastructure and licensed or open-source software tools. There are no raw materials in the manufacturing sense; the raw material of the work is data. This chapter lists the hardware and software actually used during the training, together with their specifications, approximate cost, specific use and routine maintenance.

3.2 HARDWARE USED

Table 3.1 : Hardware specifications, cost and use

Equipment	Specification	Approx. Cost (INR)	Specific Use
Development laptop	Apple silicon processor, 8-core CPU with integrated GPU, 16 GB unified memory, 512 GB SSD, macOS	1,20,000	Writing code, training the neural network models and running the web application
Secondary display	24 inch LED monitor, 1920 x 1080 resolution	12,000	Viewing code and documentation side by side
Development server	Intel Xeon processor, 32 GB RAM, 2 TB storage	2,50,000	Version control hosting and shared builds
Network switch and router	24-port Gigabit Ethernet switch with router	25,000	Local area network connectivity and internet access
Uninterruptible power supply	1 kVA online UPS with 30 minute backup	18,000	Protecting servers and workstations from power failure
External backup drive	2 TB portable hard disk drive, USB 3.0	6,000	Periodic backup of source code, datasets and trained models
Printer	Monochrome laser printer, network enabled	15,000	Printing documentation and reports

Note : The costs stated above are indicative market prices at the time of training and are given for the purpose of this report only.

3.3 SOFTWARE AND DEVELOPMENT TOOLS USED

The entire project was implemented using free and open-source software. This is a deliberate engineering choice: it removes licensing cost, it allows the work to be reproduced by anyone, and the libraries concerned are the accepted standard in the machine learning industry.

Table 3.2 : Software and tools used

Software / Tool	Version	Licence and Cost	Specific Use in the Project
Python	3.13	Open source (free)	Principal programming language for the whole project
PyTorch	2.10	Open source (free)	Deep learning framework used to define and train the LSTM network
pandas	2.3	Open source (free)	Reading, cleaning and reshaping the weather data tables
NumPy	2.3	Open source (free)	Numerical array computation and metric calculation
Matplotlib	3.10	Open source (free)	Generating the diagnostic charts saved as image files
Streamlit	1.51	Open source (free)	Building the interactive web application front end
Plotly	6.3	Open source (free)	Interactive charts inside the web
Requests	2.32	Open source (free)	Calling the Open-Meteo web service to download weather data
Git	2.39	Open source (free)	Version control of the source code
GitHub	-	Free for public repositories	Remote hosting of the source code repository
Visual Studio Code	1.9x	Free	Source code editor used for development

3.4 DATA SOURCE USED AS RAW MATERIAL

The raw material of the project is historical weather data. This was obtained from the Open-Meteo ERA5 reanalysis archive, a freely available web service that requires no registration and no access key. ERA5 is the fifth generation atmospheric reanalysis produced by the European Centre for Medium-Range Weather Forecasts, and it provides a physically consistent record of past weather conditions on a global grid.

For each of the six cities, thirty-five years of daily records were downloaded, covering the period from 1st January 1990 to 31st December 2024. This amounts to 12,784 daily records per city, each record containing fourteen weather variables. The cost of this data is nil.

3.5 ROUTINE MAINTENANCE CARRIED OUT

3.5.1 Hardware maintenance

- Weekly cleaning of workstations, keyboards and display screens.
- Monthly inspection of cable connections and power supply points.
- Quarterly testing of the uninterruptible power supply battery backup.
- Periodic checking of free disk space on the development machines.

3.5.2 Software maintenance

- Regular application of operating system and security updates.
- Updating of Python library versions after testing for compatibility.
- Weekly backup of source code, datasets and trained model files.
- Clearing of the temporary download cache when disk space is required.
- Periodic retraining of the models as newer weather data becomes available.

CHAPTER 4

PROCESSES, METHODOLOGIES AND DATA HANDLING

PROCEDURES

4.1 SOFTWARE DEVELOPMENT METHODOLOGY

The organization follows an iterative and incremental development methodology derived from Agile practice. Work is divided into short cycles, and at the end of each cycle a working increment of the system is demonstrated to the mentor and revised according to the feedback received. This approach was well suited to a machine learning project, where the accuracy of a model cannot be predicted in advance and repeated experiment is unavoidable.

4.2 THE MACHINE LEARNING PIPELINE

The complete process followed in this project consists of eight stages, carried out in sequence. Each stage is implemented as a separate Python module so that any stage may be re-executed independently.

Table 4.1 : Stages of the machine learning pipeline

Stage	Process	Description of the Operation
1	Data acquisition	Downloading 35 years of daily weather records from the Open-Meteo archive web service
2	Data cleaning	Sorting records by date, checking for missing days and filling any gaps by interpolation
3	Feature engineering	Deriving ten additional predictive columns from the fourteen raw weather variables
4	Splitting and scaling	Dividing the record chronologically into training, validation and test periods and standardizing the inputs
5	Windowing	Converting the daily series into overlapping sequences of thirty consecutive days
6	Model training	Training the LSTM network and retaining the parameters that perform best on validation data
7	Calibration	Correcting the systematic compression of predicted rainfall using validation data only
8	Evaluation and deployment	Measuring accuracy on the unseen test period and publishing the model through a web application

4.3 DATA ACQUISITION PROCEDURE

Weather data is obtained by sending requests to the Open-Meteo archive web service over the internet. A request for thirty-five years of hourly data in a single call is very large, and the service rejects such requests with an error indicating that the permitted rate has been exceeded. The acquisition procedure was therefore designed with three safeguards.

4.3.1 Division into blocks

The requested period is divided into blocks. Daily variables are requested in blocks of five years and hourly variables in blocks of two years, since hourly data is approximately twenty-four times larger for the same period.

4.3.2 Caching of downloaded blocks

Every block that is successfully downloaded is written to a cache directory on the local disk. If the procedure is interrupted, a subsequent run reads the cached blocks from disk and requests

only those that are still missing. The download is therefore resumable and no completed work is lost.

4.3.3 Controlled retry on refusal

When the service refuses a request because the permitted rate has been exceeded, the procedure waits and retries rather than failing. The waiting period is doubled after each unsuccessful attempt, beginning at twenty seconds. If the service specifies its own waiting period, that value is used instead.

4.4 FEATURE ENGINEERING

The raw data contains fourteen measured weather variables. Ten further columns were derived from these, giving twenty-four inputs in total. Each derived column encodes knowledge about how rainfall behaves, and thereby assists the network in learning. Every derived column is computed using only information available up to and including the day in question, so that no information about the future can reach the model.

Table 4.2 : Derived features and their purpose

Derived Feature	Method of Computation	Reason for Inclusion
Day-of-year sine and cosine	Sine and cosine of the day number scaled over the year	Represents the annual monsoon cycle as a smooth pair, so that 31 December and 1 January are treated as adjacent
Logarithm of rainfall	Natural logarithm of one plus the rainfall	Reduces the extreme skew of rainfall data
Rolling means over 3, 7 and 30 days	Average rainfall over the preceding period	Rainfall is strongly persistent; wet days occur in clusters
Wet streak over 7 days	Count of rainy days in the preceding week	Indicates whether a wet spell is already in progress
Pressure change	Difference in mean surface pressure from the previous day	Falling atmospheric pressure commonly precedes rainfall
Humidity change	Difference in mean relative humidity from the previous day	Rising humidity commonly precedes rainfall
Dew point spread	Mean temperature minus mean dew point	A small spread indicates that the air is close to saturation

4.5 DATA HANDLING AND STORAGE PROCEDURES

- Downloaded weather records are stored as comma separated value files, one file per city, within a dedicated data directory.
- Intermediate download blocks are held in a separate cache directory that is excluded from version control.
- Trained model parameters are stored as PyTorch checkpoint files, one per city, each containing the network weights, the scaling constants and the calibration constants.
- Evaluation results are written as structured JSON files and diagnostic charts as image files.
- All source code is placed under Git version control and stored in a remote repository on GitHub.

4.6 MODEL ARCHITECTURE AND TRAINING PROCEDURE

The network consists of a two-layer LSTM with ninety-six hidden units per layer and a dropout rate of 0.25 between layers. It receives a sequence of thirty consecutive days, each described by twenty-four features. The hidden state produced at the final time step summarizes the whole window and is passed to two separate output heads.

The first head produces the predicted quantity of rainfall, expressed on a logarithmic scale. The second head produces the probability that rainfall will exceed one millimetre. Two heads are used because daily rainfall is zero-inflated: approximately two-thirds of days are dry, and a network trained only to predict a quantity learns to protect itself by predicting values close to zero. Training a classifier alongside gives the shared network a clearer signal for the onset of a wet spell.

Table 4.3 : Model and training configuration

Parameter	Value	Reason for the Choice
Network type	Long Short-Term Memory (LSTM)	Designed to retain information across long input sequences
Number of layers	2	Sufficient depth without excessive overfitting on the available data
Hidden units per layer	96	Balances model capacity against training time
Dropout	0.25	Regularization to reduce overfitting
Input window length	30 days	Covers a complete synoptic weather cycle
Forecast horizon	1 day	Next-day prediction
Optimizer	AdamW, learning rate 0.001	Adaptive optimization with weight decay
Batch size	64	Stable gradient estimates within available memory
Gradient clipping	Maximum norm 1.0	Prevents unstable updates in recurrent networks
Early stopping patience	15 epochs	Training stops when validation performance ceases to improve

4.6.1 The loss function

The loss function combines a Huber loss on the logarithm of rainfall with a binary cross-entropy loss on the occurrence of rainfall. Three specific measures were incorporated to overcome the imbalance in the data.

- The target is transformed by the logarithm, because raw rainfall ranges from zero to 251 millimetres and would otherwise be dominated by a few extreme days.
- A positive class weight is applied to the classification loss, without which the classifier could obtain sixty-six per cent accuracy simply by always predicting dry weather.
- An intensity weight is applied to the regression loss, so that heavy rainfall days are not drowned out by the large number of ordinary dry days.

CHAPTER 5

TESTING OF SOFTWARE AND DATA HANDLING PROCEDURES

5.1 NECESSITY OF TESTING

In a machine learning project, testing has two distinct aspects. The first is conventional software testing, which verifies that the program executes correctly and produces output in the expected form. The second, and more important, is the statistical evaluation of the model, which establishes whether its predictions are actually accurate on data it has never seen. A program may run without any error and still produce a model of no value.

5.2 TESTING OF DATA INTEGRITY

Before any model was trained, each downloaded dataset was tested for completeness. The tests applied and the results obtained are given below.

Table 5.1 : Data integrity test results

Test Applied	Method	Result Obtained
Record count	Counting rows between the first and last date	12,784 records per city, as expected for 35 years
Missing dates	Comparing against a complete calendar date range	Zero missing days in every dataset
Missing values	Counting empty cells in all columns	Zero empty cells in every dataset
Duplicate dates	Checking for repeated date values	No duplicates found
Value range	Inspecting minimum and maximum of each column	All values within physically plausible limits

5.3 METHOD OF MODEL EVALUATION

The dataset was divided strictly by date, and never at random. A random division would allow the model to be trained on days occurring after the days on which it is tested, which would allow information about the future to influence the result and would produce accuracy figures

that are impressively high but entirely misleading. The division used is fixed in the source code so that the result cannot change if additional data is later downloaded.

Table 5.2 : Division of the dataset

Division	Period	Number of Samples	Purpose
Training	1990 to 2014	9,101	Learning the network parameters
Validation	2015 to 2019	1,826	Selecting the stopping point, the decision threshold and the calibration constants
Testing	2020 to 2024	1,827	Final measurement of accuracy on unseen data

The scaling constants used to standardize the inputs were computed from the training period alone. The decision threshold that separates a forecast of rain from a forecast of no rain was selected on the validation period alone. Selecting it on the test period would amount to marking one's own answer paper.

5.4 COMPARISON AGAINST BASELINES

An accuracy figure has no meaning in isolation. Two reference methods were therefore implemented, and the model must be shown to be superior to both.

5.4.1 Persistence

The persistence baseline predicts that tomorrow's rainfall will equal today's. Because rainfall is strongly correlated from one day to the next, this simple rule is surprisingly effective and forms the honest standard against which any weather model must be judged.

5.4.2 Climatology

The climatology baseline predicts the long-term average rainfall of the training period for every day, taking no account of current conditions.

5.5 EVALUATION METRICS EMPLOYED

Table 5.3 : Evaluation metrics and their meaning

Metric	Applies To	Meaning
Root Mean Square Error	Rainfall quantity	Square root of the mean squared error, in millimetres; penalizes large errors heavily
Mean Absolute Error	Rainfall quantity	Average magnitude of the error, in millimetres
Coefficient of determination	Rainfall quantity	Proportion of the variation in rainfall explained by the model; a negative value indicates performance worse than predicting the mean
Accuracy	Rain or no rain	Proportion of days classified correctly
Precision	Rain or no rain	Of the days forecast as rainy, the proportion that were actually rainy
Recall	Rain or no rain	Of the days that were actually rainy, the proportion correctly forecast
F1 score	Rain or no rain	Harmonic mean of precision and recall
Critical Success Index	Rain or no rain	Standard meteorological score excluding correctly forecast dry days
Heidke Skill Score	Rain or no rain	Skill of the forecast relative to what chance alone would achieve
Area under the ROC curve	Rain or no rain	Ability of the model to rank rainy days above dry days

5.6 TEST RESULTS OBTAINED

The results obtained on the unseen test period of 2020 to 2024 are given in Table 5.4 and Table 5.5. Every trained city exceeds its own persistence baseline on both tasks.

Table 5.4 : Rain or no rain classification results

City	Accuracy	Precision	Recall	F1	F1 (persistence)	ROC-AUC
Mumbai	0.950	0.934	0.928	0.931	0.922	0.988
Pune	0.906	0.855	0.855	0.855	0.836	0.961
Chennai	0.802	0.727	0.830	0.775	0.737	0.886
Bengaluru	0.836	0.777	0.808	0.792	0.765	0.904
Delhi	0.869	0.660	0.759	0.706	0.647	0.901

Table 5.5 : Rainfall quantity prediction results

City	Mean Annual Rainfall (mm)	RMSE (mm)	MAE (mm)	R-squared	R-squared (persistence)
Mumbai	2,455	12.24	3.90	0.462	0.201
Pune	1,141	5.91	2.08	0.343	0.111
Chennai	1,549	10.60	3.31	0.211	-0.245
Bengaluru	1,154	5.57	2.36	0.178	-0.028
Delhi	819	6.96	1.92	0.183	-0.225

It is significant that the persistence baseline records a negative coefficient of determination for Chennai, Bengaluru and Delhi. This means that in those cities, repeating today's rainfall is actually worse than simply predicting the long-run average every day. The trained model is clearly superior to both references in every case.

5.7 ERROR ANALYSIS BY RAINFALL INTENSITY

A single overall error figure can conceal important weaknesses. The test results for Mumbai were therefore broken down according to the intensity of rainfall actually observed, as shown in Table 5.6.

Table 5.6 : Error by rainfall intensity band (Mumbai)

Intensity Band	Number of Days	Mean Observed (mm)	Mean Predicted (mm)	MAE (mm)
Dry, below 1 mm	1,145	0.05	0.32	0.29
Light, 1 to 10 mm	353	4.42	6.31	3.51
Moderate, 10 to 35 mm	229	19.58	16.37	9.46
Heavy, 35 to 65 mm	73	46.91	27.60	20.49
Very heavy, above 65 mm	27	102.05	31.93	70.12

This analysis reveals the principal limitation of the system, which is reported here rather than concealed. The timing of rainfall is predicted well, but its magnitude is compressed: on the twenty-seven days of very heavy rainfall, the model predicted an average of 31.93 millimetres against an observed average of 102.05 millimetres. Part of this shortfall cannot be removed,

because an intense cloudburst is produced by small-scale atmospheric convection that is simply not present in daily average measurements taken at a single location.

5.8 FUNCTIONAL TESTING OF THE APPLICATION

Table 5.7 : Functional test cases

No.	Test Case	Expected Result	Status
1	Launch the web application	Application opens and displays the forecast page	Pass
2	Select each city in turn	Correct city model, data and metrics are loaded	Pass
3	Select a city having no trained model	A clear message is displayed with the command to train it; no incorrect forecast is shown	Pass
4	Request a live forecast	Data is fetched up to the current date and a forecast for the next day is displayed	Pass
5	Disconnect the network and request a live forecast	Application reports that live data is unavailable and reverts to stored data	Pass
6	Interrupt the data download midway	Completed blocks are retained and the next run resumes from that point	Pass
7	Run the download when data already exists	No network request is made and the existing dataset is retained	Pass
8	View the model performance page	Metric tables and diagnostic charts are displayed correctly	Pass

CHAPTER 6

SAFETY PROCEDURES AND SAFETY MEASURES FOLLOWED

6.1 INTRODUCTION

In a software organization the hazards are different from those of a manufacturing plant. There are no heavy machines, moving parts or hazardous chemicals. The risks arise instead from electrical equipment, from fire, from prolonged work at a computer, and from the loss or disclosure of data. Safety procedures are accordingly directed at these four areas.

6.2 ELECTRICAL SAFETY

- All equipment is connected through properly earthed three-pin power outlets.
- Servers and workstations are protected by uninterruptible power supply units against sudden power failure.
- Power cables are routed through trunking and are not left across walkways where they may be damaged or cause a person to trip.
- Extension boards are not overloaded, and multiple boards are never connected in series.
- Only authorized personnel of the system administration team open equipment cabinets.

6.3 FIRE SAFETY

- Carbon dioxide fire extinguishers, which are suitable for electrical fires, are provided on every floor and near the server room.
- Extinguishers are inspected periodically and the date of inspection is recorded on the attached tag.
- Emergency exits and escape routes are clearly marked and are kept free of obstruction at all times.

- Fire drills are conducted and the assembly point outside the building is known to all staff.
- Smoke detectors are installed in the server room, which is also kept air-conditioned to prevent overheating.

6.4 OCCUPATIONAL HEALTH AND ERGONOMIC PRACTICE

Continuous work at a computer can cause eye strain and disorders of the back, neck and wrists.

The following practices were observed during the training period.

- Adjustable chairs with adequate lower back support are provided, and the display is positioned at approximately eye level.
- A short break is taken away from the screen after every hour of continuous work.
- Adequate lighting is maintained and screen glare is reduced by suitable positioning of the workstations.
- The keyboard and mouse are placed so that the wrists remain in a neutral position.

6.5 DATA SAFETY AND INFORMATION SECURITY

The protection of data and of access credentials is treated as a safety matter, because their loss or disclosure can cause serious damage. The following practices were followed and were applied directly in this project.

6.5.1 Protection of access credentials

- Passwords and access tokens are never written into the source code.
- Credentials are never sent through electronic mail or chat messages in plain text.
- Any credential that is accidentally exposed is revoked immediately and a new one is generated.

- Before the source code was published, it was searched to confirm that no key or token had been included.

6.5.2 Protection of data

- Source code, datasets and trained models are backed up weekly to an external drive.
- All source code is maintained under version control, so that any earlier state can be recovered.
- Access to the server room is restricted and the computers are protected by passwords with automatic screen locking.
- Files that need not be published, such as temporary caches, are explicitly excluded from the version control system.

6.6 SAFETY EQUIPMENT AVAILABLE AT THE WORKPLACE

Table 6.1 : Safety equipment provided

Safety Equipment	Purpose
Carbon dioxide fire extinguisher	Extinguishing electrical fires without damaging equipment
Smoke detector and alarm	Early detection of fire in the server room
Uninterruptible power supply	Preventing data loss and hardware damage during power failure
Earthed power distribution	Protecting personnel from electric shock
First-aid box	Immediate treatment of minor injuries
Anti-static wrist strap	Preventing electrostatic damage while handling internal computer components
Emergency exit signage	Guiding safe evacuation of the premises

CHAPTER 7

PARTICULARS OF PRACTICAL EXPERIENCE IN THE ORGANIZATION

7.1 INTRODUCTION

This chapter records the practical experience gained during the training period. In a software organization the equivalents of production, assembly, testing and maintenance are respectively the development of program modules, their integration into a working system, the verification of that system, and its continued upkeep after delivery.

7.2 EXPERIENCE IN DEVELOPMENT

The trainee was engaged in writing program modules in the Python language. The work included reading data from a web service, transforming tabular data, defining a neural network in the PyTorch framework, and writing training and evaluation routines. Emphasis was placed by the mentor on writing code that another engineer can read: descriptive names, short functions with a single purpose, and comments that explain why a decision was taken rather than restating what the code does.

7.3 EXPERIENCE IN INTEGRATION

Individual modules were combined into a single pipeline in which the output of one stage becomes the input of the next. A shared module holding the list of cities and the derived file paths was introduced so that the same information is not written in several places. This experience demonstrated the practical value of keeping a single authoritative definition of any item of information.

7.4 EXPERIENCE IN TESTING

Testing occupied a substantial part of the training. Beyond checking that the program executed, it was necessary to establish that the model was genuinely accurate. Comparison against baseline methods, examination of results separately for each intensity of rainfall, and careful avoidance of any leakage of future information were all practised. The most important lesson learnt was that a favourable overall figure may conceal a serious weakness, which becomes visible only when the results are examined in detail.

7.5 EXPERIENCE IN DEPLOYMENT

The completed model was deployed through a web application written using the Streamlit framework, which allows a Python program to be presented as an interactive web page. Preparing an application for deployment on a hosting service, and reducing the size of its dependencies so that it would fit within the permitted limit, formed part of this experience.

7.6 EXPERIENCE IN MAINTENANCE

Maintenance activities included correcting defects discovered after a module had been declared complete, updating documentation to match the code, retraining models when the data or the loss function changed, and managing the source code repository so that the history of changes remained clear and meaningful.

7.7 NON-TECHNICAL EXPERIENCE

- Participating in daily stand-up meetings and reporting progress concisely.
- Accepting and acting upon review comments on submitted work.
- Maintaining a daily record of work performed and difficulties encountered.
- Observing office discipline, punctuality and professional conduct.
- Presenting completed work to the mentor and answering questions on it.

CHAPTER 8

DETAILED REPORT OF THE TASKS UNDERTAKEN

8.1 OVERVIEW OF THE WORK SCHEDULE

The training was organized into weekly assignments. Table 8.1 summarizes the work carried out in each week, and the sections that follow describe the principal tasks in detail.

Table 8.1 : Week-wise schedule of work

Week	Task Undertaken	Outcome
1	Study of the problem, of machine learning fundamentals and of the LSTM architecture; setting up the development environment	Development environment ready; approach finalized
2	Development of the data acquisition module for the Open-Meteo archive	35 years of daily weather data downloaded for Mumbai
3	Data cleaning, feature engineering and construction of input sequences	24 input features prepared; sequences of 30 days constructed
4	Design of the two-headed LSTM network and the loss function	Model defined in PyTorch and training commenced
5	Training, hyper-parameter experiments and error analysis	Working model obtained; weakness in heavy rainfall identified
6	Development of the calibration procedure and re-evaluation	Coefficient of determination improved from 0.327 to 0.462
7	Development of the web application using Streamlit	Interactive application with four pages completed
8	Extension of the system to six cities and per-city models	Five city models trained; comparison page added
9	Implementation of real-time forecasting	Forecasts for the current date made available
10	Documentation, version control and preparation of this report	Source code published; report prepared

8.2 TASK 1 : DEVELOPMENT OF THE DATA ACQUISITION MODULE

A Python module was written to download daily weather records from the Open-Meteo archive web service. Ten variables are available directly as daily values; four further variables, namely relative humidity, surface pressure, cloud cover and dew point, are available only at hourly

intervals and were averaged to daily values within the module. The block division, disk cache and controlled retry described in Chapter 4 were all implemented in this task.

8.3 TASK 2 : FEATURE ENGINEERING AND SEQUENCE CONSTRUCTION

The fourteen raw variables were extended to twenty-four by the addition of the derived features listed in Table 4.2. The daily series was then converted into overlapping windows of thirty consecutive days, each window being paired with the rainfall of the following day as its target. Particular care was taken to ensure that no feature could contain information about the day being predicted.

8.4 TASK 3 : DESIGN OF THE NETWORK AND LOSS FUNCTION

The two-headed LSTM described in Chapter 4 was implemented. The combined loss function was written as a separate class so that the weight given to each component could be varied during experimentation.

8.5 TASK 4 : TRAINING AND HYPER-PARAMETER EXPERIMENTS

The models were trained on the graphics processor of the development machine, each city requiring approximately fifty seconds. A systematic experiment was conducted on the intensity weighting of the loss function, in which six configurations were trained and compared. The configuration was selected on the validation period alone, never on the test period. The selected configuration improved the validation coefficient of determination from 0.304 to 0.356.

8.6 TASK 5 : DEVELOPMENT OF THE CALIBRATION PROCEDURE

Error analysis showed that predicted rainfall was systematically compressed towards the middle of its range, so that heavy rainfall was consistently under-predicted. This is a known consequence of training with a loss function that seeks an average value. A linear correction

was therefore fitted on the logarithmic scale, using the validation period only, of the form given below.

$$\log(1 + \text{corrected rainfall}) = 1.201 \times \log(1 + \text{predicted rainfall}) - 0.104$$

A slope greater than one stretches the compressed distribution of forecasts outward. This single measure produced the largest improvement in the project, raising the coefficient of determination on the test period from 0.327 to 0.462 and reducing the systematic bias from -3.41 to -1.67 millimetres. The correction is legitimate precisely because its constants are computed without any reference to the test data.

8.7 TASK 6 : EVALUATION AGAINST BASELINES

An evaluation module was written to compute all ten metrics listed in Table 5.3, to compare the model against the persistence and climatology baselines, and to break the error down by intensity of rainfall. The results are written to a structured file and a set of six diagnostic charts is generated.

8.8 TASK 7 : DEVELOPMENT OF THE WEB APPLICATION

An interactive web application was developed using the Streamlit framework. It presents five pages, described in Table 8.2.

Table 8.2 : Pages of the web application

Page	Function
Forecast	Displays the probability of rainfall and the expected quantity for the following day, together with the thirty-day window supplied to the model
History explorer	Compares predicted and observed rainfall for any selected year, restricted by default to years not used in training
Model performance	Presents the test metrics, the comparison with baselines and the breakdown by intensity
Compare cities	Presents all trained city models side by side, each against its own persistence baseline
How it works	Explains the design of the loss function, the calibration step and the precautions taken against leakage

8.9 TASK 8 : EXTENSION TO MULTIPLE CITIES

The system was extended from one city to six. A separate model, with its own scaling constants, calibration constants and decision threshold, is trained for each city. Sharing a single model across cities was considered and rejected, for the reason given in Section 10.3. A driver program was written to train and evaluate every city in sequence and to produce a comparative summary.

8.10 TASK 9 : IMPLEMENTATION OF REAL-TIME FORECASTING

The ERA5 archive used for training lags real time by approximately five days, and therefore cannot answer a question about tomorrow. A second service, the Open-Meteo forecast endpoint, provides the same variables up to the current date. A module was written to obtain a window ending on the current date from this service, and the web application was modified to offer a choice between live and stored data. This service maintains a separate usage limit, so live forecasting continues to function even when the archive service is temporarily refusing requests.

8.11 TASK 10 : DOCUMENTATION AND VERSION CONTROL

The complete source code was placed under Git version control and published to a remote repository. A detailed documentation file was prepared describing the installation procedure, the results obtained, the known limitations and the possibilities for future work. Files that need not be published were explicitly excluded, and the repository was searched to confirm that no access credentials had been included.

CHAPTER 9

SPECIAL AND CHALLENGING EXPERIENCES DURING TRAINING

9.1 INTRODUCTION

Several difficulties were encountered during the training that were not anticipated at the outset. Each required investigation and a considered solution, and each taught a lesson of general application. They are described in this chapter together with the manner in which they were resolved.

9.2 REFUSAL OF SERVICE BY THE DATA PROVIDER

The most persistent practical difficulty was the refusal of the weather service to accept requests once a usage limit had been exceeded, signalled by the error code 429. The first version of the download program waited only a few seconds before retrying, and worse, discarded all previously downloaded blocks when the final block failed. Six successful downloads were lost because the seventh was refused.

The program was redesigned so that every block is cached on disk as soon as it is received, so that the waiting period doubles after each refusal, and so that the failure of one city does not abandon the remaining cities. This experience taught the importance of designing a program to expect and survive failure of an external service, rather than assuming that it will always respond.

9.3 IMBALANCE IN THE DATA

Approximately two-thirds of the days in the record are dry. A classifier can therefore obtain sixty-six per cent accuracy by predicting dry weather on every day, without learning anything at all. This was overcome by applying a positive class weight to the classification loss, and it

taught the lesson that accuracy alone is an unreliable measure when the classes are unequal in size.

9.4 SYSTEMATIC UNDER-PREDICTION OF HEAVY RAINFALL

The most instructive difficulty of the training was the discovery that the first working model, although it appeared satisfactory overall, predicted approximately fifteen millimetres of rainfall on days when more than one hundred millimetres actually fell. The overall metrics had concealed this entirely; it became visible only when the errors were examined separately for each intensity band.

Investigation showed two causes. The Huber loss function ceases to increase steeply beyond a threshold, which is precisely the range in which heavy rainfall errors lie, so those days contributed only a small gradient against the mass of ordinary dry days. In addition, a model trained to predict an average value inevitably produces a distribution narrower than reality. The first cause was addressed by intensity weighting of the loss, and the second by the calibration described in Section 8.6.

9.5 SILENT CHANGE IN THE EVALUATION PERIOD

A subtle and instructive defect was discovered during testing of the web application. A control provided to refresh the stored data extended the Mumbai dataset by two additional years. Because the test period had been defined simply as everything after a certain date, the test period silently grew from 1,827 days to 2,430 days, and the reported accuracy changed accordingly. Comparison between cities was no longer valid.

The remedy was to fix the boundaries of the training, validation and test periods explicitly in the source code, so that results cannot change merely because additional data has been downloaded. The lesson learnt is that an experimental result must be reproducible, and that anything capable of changing it silently is a defect.

9.6 DIFFICULTY IN PREPARING THE APPLICATION FOR DEPLOYMENT

When the application was prepared for deployment on a public hosting service, it was found that the standard distribution of the PyTorch library for the Linux platform includes support for graphics processors and occupies approximately 2.5 gigabytes, far exceeding the permitted size. The difficulty was resolved by specifying the processor-only distribution of the library, which occupies approximately 200 megabytes and is entirely sufficient for a network of this size.

9.7 ASPECTS OF THE WORKPLACE THAT WERE LIKED

- The freedom given to investigate a problem independently before assistance was offered.
- The insistence of the mentor that a result be verified rather than assumed.
- The practice of daily stand-up meetings, which made the progress of the whole team visible.
- The willingness of senior engineers to explain the reasoning behind their design decisions.
- Exposure to the complete lifecycle of a product, from raw data to a deployed application.

9.8 ASPECTS THAT WERE FOUND DIFFICULT

- Long periods of training and downloading during which little visible progress could be made.
- The absence of a definite criterion for when a model is good enough to be considered complete.

- The initial difficulty of accepting that a limitation of the data cannot be removed by improving the program.
- Working within the usage limits of an external service, which occasionally delayed planned work.

CHAPTER 10

CONCLUSION

10.1 SUMMARY OF THE WORK CARRIED OUT

During the industrial training a complete rainfall prediction system was designed, implemented, tested and deployed. Thirty-five years of daily weather records were obtained for six Indian cities from the Open-Meteo ERA5 archive. Twenty-four predictive features were derived from fourteen measured variables. A two-layer Long Short-Term Memory network with two output heads was trained separately for each city to predict, one day ahead, both the probability and the quantity of rainfall. The models were evaluated on a five-year period that was never used during training, and were compared against two baseline methods. An interactive web application was developed to deliver forecasts, including forecasts for the current date obtained from live data.

10.2 RESULTS ACHIEVED

- For Mumbai, the model classifies rain against no rain with an F1 score of 0.931 and an area under the ROC curve of 0.988.
- For Mumbai, the model explains 46.2 per cent of the variation in daily rainfall quantity, against 20.1 per cent for the persistence baseline.
- Every trained city exceeds its own persistence baseline on both the classification and the quantity prediction tasks.
- For Chennai, Bengaluru and Delhi the persistence baseline records a negative coefficient of determination, whereas the trained models remain clearly positive.
- The calibration procedure improved the coefficient of determination for Mumbai from 0.327 to 0.462.

- A working web application was produced, providing forecasts for the current date.

10.3 OBSERVATIONS ON MODEL BEHAVIOUR

A clear relationship was observed between the accuracy attained and the degree to which a city's rainfall is dominated by the monsoon. Mumbai receives approximately 2,455 millimetres of rain in a concentrated season from June to September, and this strong seasonal structure is precisely what a sequence model is able to exploit. Delhi and Bengaluru receive much of their rainfall from scattered convective storms, which are far less predictable from the previous days' local averages, and the accuracy attained there is correspondingly lower.

This observation also justifies the decision to train a separate model for each city rather than one shared model. The rainfall regimes of Indian cities differ fundamentally: Chennai receives most of its rain in the northeast monsoon from October to December, which is the opposite half of the year from Mumbai, while Bengaluru has two separate wet seasons. Standardizing the rainfall of one city using the average of another would distort every input before the network received it.

10.4 LIMITATIONS OF THE PRESENT SYSTEM

- The magnitude of very heavy rainfall is systematically under-predicted, as documented in Section 5.7. Forecasts above approximately twenty millimetres should be interpreted as an indication that heavy rain is likely, not as a precise quantity.
- The forecast horizon is limited to one day. Accuracy would be expected to fall as the horizon is extended.
- The model uses observations from a single grid point and therefore cannot observe a weather system approaching from a neighbouring region.
- The models are trained on reanalysis data but operate on operational analysis data when producing live forecasts, which introduces a small additional error.

10.5 SCOPE FOR FUTURE WORK

- Including weather variables from neighbouring grid points, which is the most promising means of improving the prediction of heavy rainfall.
- Predicting the tenth, fiftieth and ninetieth percentiles of rainfall instead of a single value, so that an honest range of uncertainty is provided.
- Evaluating an attention-based or transformer architecture on the same data for comparison with the LSTM.
- Extending the system to further cities and to longer forecast horizons.
- Incorporating satellite imagery or radar data, which observe convective activity directly.

10.6 CONCLUDING REMARKS

The training provided practical experience of the complete lifecycle of a machine learning product, from the acquisition of raw data to a deployed and documented application. The most valuable lessons were not concerned with the neural network itself but with engineering discipline: that an external service will fail and a program must be designed to survive it, that a favourable overall metric may conceal a serious weakness, that a result which can change silently is not a result at all, and that a model must always be measured against a simple baseline before it can be claimed to be useful. The objectives set at the beginning of the training were met, and a working system was delivered.

CHAPTER 11

REFERENCES AND SOURCES OF INFORMATION

11.1 BOOKS AND RESEARCH PAPERS

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- [5] Wilks, D. S., "Statistical Methods in the Atmospheric Sciences", Fourth Edition, Elsevier, 2019.

11.2 WEB RESOURCES

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- [7] Open-Meteo Historical Weather API, <https://open-meteo.com/en/docs/historical-weather-api>
- [8] PyTorch Documentation, <https://pytorch.org/docs/stable/index.html>
- [9] PyTorch LSTM Module Reference, <https://pytorch.org/docs/stable/generated/torch.nn.LSTM.html>
- [10] Streamlit Documentation, <https://docs.streamlit.io>
- [11] pandas Documentation, <https://pandas.pydata.org/docs>
- [12] NumPy Documentation, <https://numpy.org/doc/stable>
- [13] Plotly Python Graphing Library, <https://plotly.com/python>
- [14] India Meteorological Department, <https://mausam.imd.gov.in>

11.3 PROJECT REPOSITORY

[15] Project source code repository, <https://github.com/muradepratiksha18-bit/rain-forecasting-project>

11.4 OTHER SOURCES

- Guidance and technical discussions with the mentor and the engineering staff of the organization.
- Internal documentation and coding standards of the organization.
- Course material of the Diploma in Computer Technology, Maharashtra State Board of Technical Education.